



COMPARATIVE ANALYSIS OF PATHFINDING ALGORITHMS FOR URBAN NAVIGATION

AHMED RAGAB , AYA MOHAMED , SHAMS HESHAM , ABDELRAHMAN MAMDOUH ,
ABDELRAHMAN WAHEED
ELE335 (ARTIFICIAL INTELLIGENCE), DR: MOHAMED REHAN

PROBLEM STATEMENT

apply different search algorithms on different real map and compare between their performance

OBJECTIVE

implementing different search algorithms (A*, greedy best first, BFS and Bidirectional BFS) and apply them between two random locations on the map and compare their performance using (different heuristics (euclidean distance, time), no of nodes visited and explored)

PROBLEM FORMULATION

- States Representation: all streets in the map represent a state.
- Initial State: randomly chosen start street.
- Actions: drive in different streets.
- Transition model: move from street to another street in Qalyub.
- goal State: the randomly chosen end street is reached.
- Path Cost function: distance or time between street n and the destination street.

TECHNIQUES USED

Algorithm	Type	Key Feature
BFS	Uninformed	Shortest hops
Bidirectional BFS	Uninformed	2× faster than BFS
A*	Informed	Optimal path
Greedy Search	Informed	Fast but suboptimal

• **Tools:** Python, OSMnx, Matplotlib.

• **Heuristics:**

- Euclidean distance
- time-based.

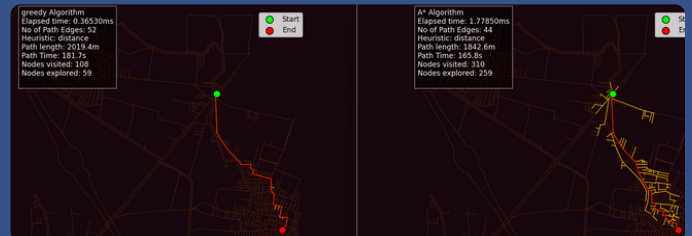
$$h(n) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$h(n) = \frac{\text{distance}(n, \text{goal})}{\text{max speed}}$$

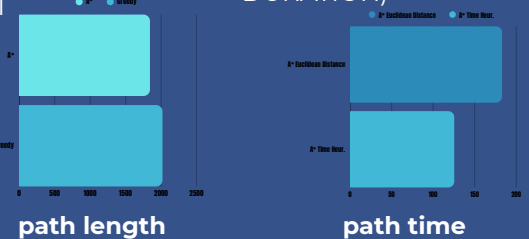
• **Animation:** Step-by-step exploration visuals.



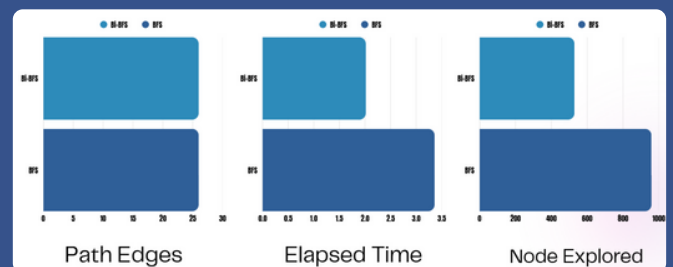
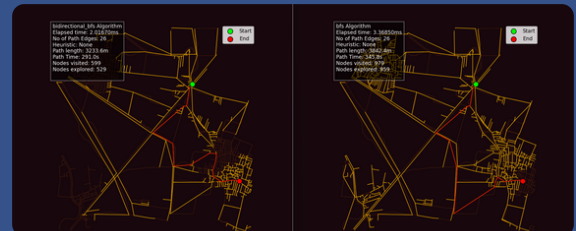
ANALYSIS



- A* always finds the optimal path in terms of path length
- greedy is often faster, but not optimal
- A* WITH TIME HEURISTICS ALWAYS FIND THE OPTIMAL PATH (LEAST PATH DURATION)



- **bfs vs bidirectional analysis**



- Bidirectional BFS is approximately twice as fast, explores fewer nodes, and visits fewer nodes
- here we don't care about path cost assumed unweighted graph
- both finds the optimal path (least no of edges)

RESULTS & RECOMMENDATIONS

- FOR OPTIMAL PATHS (SHORTEST OR LEAST TIME): USE A*.
- FOR OPTIMAL PATH (LEAST NO OF EDGES): USE BIDIRECTIONAL BFS OR GREEDY.
- AVOID GREEDY IF OPTIMALITY IS CRITICAL.